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1 Introduction

The Tile Calorimeter (TileCal) is a sampling calorimeter of the ATLAS Experiment [1]. TileCal is in charge of identifying and measuring hadronic jets and transverse missing energy. The TileCal electronics will be upgraded to face the higher radiation levels and increased rates of pileup at the High Luminosity Large Hadron Collider (HL-LHC) [2]. The upgrade project aims to provide continuous digital read-out of all the TileCal with improved timing and better energy resolution.

The off-detector electronics of the upgraded TileCal will provide digitized signals at 40 MHz to the trigger systems through the Trigger and Data Acquisition interface (TDAQi), while the so-called Front-End Link eXchange (FELIX) system will read-out data stored in pipelines in Tile Preprocessors (TilePPr) at 1 MHz. The TilePPrs will interface the on- and off-detector electronics through multi-Gbps optical links. The power will be monitored and distributed to the on-detector electronics by the Detector Control System (DCS) that interfaces with the Low Voltage Power Supplies (LVPS) in charge of distributing power to all the digital and analogue electronics, and the High Voltage Power Supplies that will provide high voltage to each PMT.

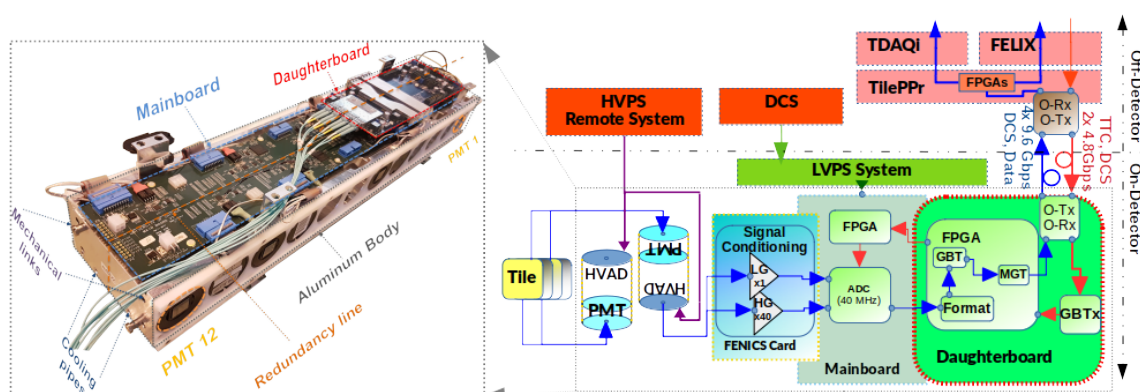


Figure 1. (Left:) TileCal Phase-II Upgrade Minidrawer. (Right:) On-Detector and Off-detector electronics block diagram.